

Noah Sun

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Education

University of Waterloo – BSc. in Mechatronics Engineering (Candidate) September 2025 – April 2030
Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic GPA: 3.92 / 4.00

Experience

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers to develop a 32 MB flash driver using C++ on the STM32 Nucleo to store config details and crash logs
- Built an RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

Forward Deployed Engineer – Ascendance Foundry January 2026 – April 2026

- Developed 5 AI workflow automations, introducing input validation and reducing human error by 80%
- Built a GTM automation that uses Python Playwright to find leads and generate outreach emails using Claude API
- Implemented 2 ETL pipelines using Azure Data Factory, delivering per-project statistic dashboards in Power BI

Co-founder and Lead Developer – Elapse | ▶ [Google Play](#) | [App Store](#) | [GitHub](#) June 2024 – August 2025

- Led 4 developers to build an iOS/Android Flutter app serving the 100k+ member VEX Robotics Competition community with real-time schedules, rankings, and match data, receiving 3500+ downloads and a 5.0-star rating
- Designed adaptive match times and scouting forms, increasing team productivity at tournaments by 35%
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by 30%

V5RC Team Programming Lead – Checkmate Robotics | ▶ [Demo Video](#) | [GitHub](#) June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the Create (innovation) Award at VEX Worlds 2024
- Engineered ArkLib, a modular C++ motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in 5 consecutive years and winning the Think (programming) Award at international events
- Designed and tuned PID and control algorithms, achieving 1"/1° precision movements across 300+ matches

Vex AI Competition Team Embedded Systems Lead – Mi3L Schools May 2024 – June 2024

- Developed an API for the VEX Brain to process 3D spatial data from the Intel RealSense depth camera via the Jetson Nano for adaptive field navigation, winning the Innovate Award and Skills World Champion at VEX AI Worlds 2024

Projects

Signal - LLM Agent Simulation Platform | [GitHub](#)

- Built a multi-layer LLM agent simulation and dashboard in Python and JavaScript with clean separation across a simulation kernel, Claude tool-use harness, and FastAPI/SSE server
- Implemented structured tool-use over Claude API with prompt caching, reducing per-step latency and token cost

CityWatch | [GitHub](#)

- Led a team of 4 to build a reporting platform using React Native to analyze 50k+ reports to identify high-risk zones
- Trained a MobileNetV2 road defect classifier with OpenCV preprocessing pipeline, achieving 95%+ accuracy
- Forecasted 30-365 day risk heatmaps from spatial incident patterns and features stored in a PostGIS database

VexDex | [GitHub](#)

- Built full-stack competition data analysis with FastAPI endpoints, managing 8k+ data points over 40+ metrics
- Engineered composite-key relational schema with indexing in Azure SQL, reducing database query time to <100ms

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart

Frameworks/AI: ROS2, OpenCV, TensorFlow, Claude API, FastAPI, Express.js, Node.js, React Native, Flutter, Matplotlib

Data/Backend: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas

Developer Tools: Git, Docker, Linux, GitHub Actions, Foxglove, REST APIs, SSE, Microsoft Power Apps, MS Office

Hardware: Raspberry Pi, STM32 Nucleo, NVIDIA Jetson Nano, Arduino